Step 5B: Supervised PCA Comparison

Supervised PCA first screens variables by their relationship with turnout, then runs ordinary PCA on the screened variable set. It is less directly supervised than PLS, but easier to explain: turnout chooses the variable set; PCA summarizes the selected predictors.

Model Comparison

Model	Predictors/Screened Vars	Components	CV R2	CV RMSE
Full unfiltered PLS	70	12	0.565	0.057
Step 3 cleaned PLS	27	5	0.558	0.058
Supervised PCA	15	8	0.540	0.059

Best screen: corr_ge_0.15.

Largest PCA Loadings In Best Screen

Variable	Max	loading	
block5_ksi_collision_events_2021_2025_per_1000	0.633		
block1_age_65_plus_share	0.630		
block4_mayoral_top_two_margin	0.554		
block1_bachelors_or_higher_25_64_share	0.513		
block5_social_housing_share	0.498		
block4_provincial_margin	0.488		
block1_low_income_lim_at_share	0.434		
block4_effective_mayoral_candidates_5pct	0.430		
block5_no_car_household_share	0.403		
block3_visible_minority_share	0.402		
block4_federal_margin	0.401		
block3_non_citizen_share	0.396		
block3_recent_immigrant_share	0.389		
block5_school_age_5_17_share	0.358		
block1_average_household_size	0.355		

Interpretation

Supervised PCA is a useful comparison because it asks whether a simple turnout-screened latent structure can compete with PLS. If it performs similarly, the final story can lean more on interpretable screened dimensions. If it underperforms, PLS remains the stronger supervised reduction method.

The best supervised PCA screen retained these plain-language variables: visible minority share, bachelor or higher education share, effective mayoral candidates above five percent, average household size, non citizen share, recent immigrant share, mayoral top two margin, federal margin, low income share, school age children share, social housing share, provincial margin, age 65 plus share, no car household share, KSI collisions per 1000 residents. This model is useful as a reference-category check because it independently returns a compact set around racialized and citizenship geography, education and class, household structure, electoral competitiveness, age structure, social housing, carlessness, and KSI collisions. Because its predictive performance is weaker than PLS, I would use it as supporting evidence rather than the main model.